

Unit 03: Data Communication

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Self Assessment

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Objectives

After studying this unit, you will be able to:

- Discuss data communication
- Learn about the local and global reach of network
- Explain data communication with standard telephone lines
- Explain data communication with modems
- Understand data communication using digital data connections
- Explain wireless networks

Introduction

Data Communications is the transfer of data or information between a source and a receiver. The source transmits the data and the receiver receives it. The actual generation of the information is not part of data communications nor is the resulting action of the information at the receiver. Data Communication is interested in the transfer of data, the method of transfer, and the preservation of the data during the transfer process.

3.1 Data Communication

In Local Area Networks, we are interested in “connectivity”, connecting computers to share resources. Even though the computers can have different disk operating systems, languages, cabling, and locations, they still can communicate with one another and share resources.

The purpose of Data Communications is to provide the rules and regulations that allow computers with different disk operating systems, languages, cabling, and locations to share resources. The rules and regulations are called protocols and standards in data communications.

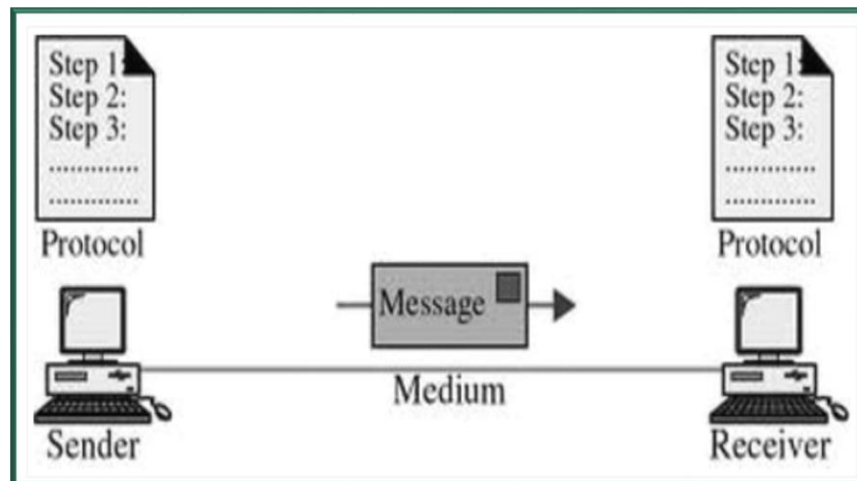
Effectiveness of a Data Communication System

The effectiveness of a data communication system depends on the three fundamental characteristics:

1. **Delivery:** System must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
2. **Accuracy:** System must deliver data accurately. Data that have been altered in transmission and left uncorrected are rustles.
3. **Timelines:** The system must deliver data on time. The data that are delivered late are useless. For example: In the case of video, audio, and voice data, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. The timely delivery of data is called real-time transmission.

Components of a Data Communication System

As we have discussed the effectiveness parameters for a data communication system. Let us now understand the basic requirements for a working communication system. Initially, the sender/source creates a message to be transmitted where a medium carries the message. The receiver (sink) who receives the message. The data communication system constitutes of the following components:



1. **Message:** A message in its most general meaning is an object of communication. It is a vessel that provides information. It can also be this information. Therefore, its meaning is dependent upon the context in which it is used; the term may apply to both the information and its form.
2. **Sender:** The sender will have some kind of meaning she wishes to convey to the receiver. It might not be conscious knowledge, it might be a subconscious wish for communication. What is desired to be communicated would be some kind of idea, perception, feeling, or datum.
3. **Receiver:** These messages are delivered to another party. The other party also enters into the communication process with ideas & feelings that will undoubtedly influence their understanding of your message and their response.
4. **Medium:** The medium acts as a means to exchange/transmit the message. The sender must choose an appropriate medium for transmitting the message else the message might not be conveyed to the desired recipients. The choice of an appropriate medium of communication is essential for making the message effective and correctly interpreted by the recipient. However, the choice of communication medium varies depending upon the features of communication.

5. **Protocols:** Protocols implies the formal description of digital message formats and the rules for exchanging those messages in or between computing systems and telecommunications. The protocols may include signaling, authentication, and error detection and correction syntax, semantics, and synchronization of communication and may be implemented in hardware or software, or both.
6. **Feedback:** Feedback is the main component of the communication process that permits the sender to analyze the efficacy of the message and confirm the correct interpretation of the message by the decoder. It may be verbal (through words) or non-verbal (in form of smiles, sighs, etc.) and may take written form also in form of memos, reports, etc.

3.2 Local and Global Reach of the Network

Data transmission, digital transmission, or digital communications is the physical transfer of data (a digital bitstream) over a point-to-point or point-to-multi-point communication channel. Examples of such channels are Copper wires, optical fibers, wireless communication channels, and storage media. The data is represented as an electromagnetic signal, such as an electrical voltage, radio waves, microwaves, or infrared signals.

The messages are either represented by a sequence of pulses using a line code (baseband transmission), or by a limited set of continuously varying waveforms (passband transmission), using a digital modulation method. The passband modulation and corresponding demodulation (also known as detection) are carried out by modem equipment. According to the most common definition of digital signal, both baseband and passband signals representing bit-streams are considered as digital transmission, while an alternative definition only considers the baseband signal as digital, and passband transmission of digital data as a form of digital-to-analog conversion.

The data transmitted may be digital messages originating from a data source (a computer or a keyboard). It may also be an analog signal such as a phone call or a video signal, digitized into a bit-stream for example using pulse-code modulation (PCM) or (analog-to-digital conversion and data compression) schemes. However, this source coding and decoding are carried out by code equipment.

Computer networking/ Data communications (Datacom) is the engineering discipline concerned with the communication between computer systems or devices. It is a sub-discipline of telecommunications, computer science, IT, and/or computer engineering that relies heavily upon the theoretical & practical application of these scientific and engineering disciplines. A computer network is any set of computers or devices connected with the ability to exchange data. There are three types of networks: Internet, intranet, and extranet.

3.3 Computer Networks

A computer network, often simply referred to as a network, is a collection of computers and devices interconnected by communications channels that facilitate communications and allows sharing of resources and information among interconnected devices.

Networking Methods

The examples of different networking methods are:

- Local area network (LAN), which is usually a small network constrained to a small geographic area. An example of a LAN would be a computer network within a building.
- Metropolitan area network (MAN), which is used for medium size areas. examples for a city or a state.
- Wide area network (WAN) is usually a larger network that covers a large geographic area.
- Wireless LANs and WANs (WLAN & WWAN) are the wireless equivalents of the LAN and WAN.

One way to categorize computer networks is by their geographic scope, although many real-world networks interconnect Local Area Networks (LAN) via Wide Area Networks (WAN) and wireless wide area networks (WWAN). All the networks are interconnected to allow communication with a variety of different kinds of media, including twisted-pair copper wire cable, coaxial cable, optical

fiber, power lines, and various wireless technologies. The devices can be separated by a few meters (e.g. via Bluetooth) or nearly unlimited distances (e.g. via the interconnections of the Internet). Networking, routers, routing protocols, and networking over the public Internet have their specifications defined in documents called RFCs. Let us explore each in detail:



Figure 1: A LAN Network

Local Area Network (LAN): A small network constrained to a small geographic area (Figure 1: A LAN Network). It spans a relatively small space and provides services to a small number of people. A set of arbitrarily located users share a set of servers, and possibly also communicate via peer-to-peer technologies. Example of LAN includes A computer network within a building. The users can share printers and some servers from a workgroup, which usually means they are in the same geographic location and are on the same LAN, whereas a network administrator is responsible to keep that network up and running.

Client-Server and Peer-to-Peer Network: A peer-to-peer or client-server method of networking methods can be used (Figure 2). A peer-to-peer network is where each client shares their resources with other workstations in the network. Examples of peer-to-peer networks are small office networks where resource use is minimal and a home network.

In a client-server network, every client is connected to the server and each other. The client-server networks use servers in different capacities. These can be classified into two types: Single-service servers and print servers. The server performs one task such as a file server, while other servers can not only perform in the capacity of file servers and print servers but also can conduct calculations and use them to provide information to clients (Web/Intranet Server). Computers may be connected in many different ways, including Ethernet cables, Wireless networks, or other types of wires such as power or phone lines.

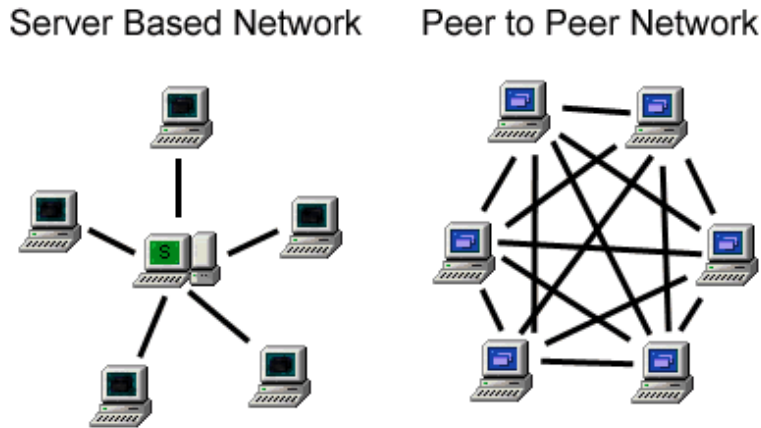


Figure 2: Client-Server & Peer-to-Peer Network

Metropolitan Area Network (MAN): It is used for medium size areas. MAN is a computer network that interconnects users with computer resources in a geographic region of the size of a metropolitan area (Figure 3). Examples: Network for a city or a state.

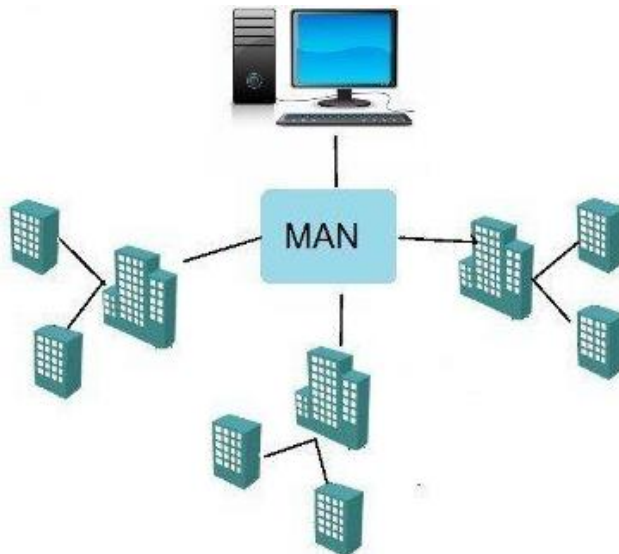


Figure 3: A Metropolitan-Area Network

Wide Area Network (WAN): A WAN is a network where a wide variety of resources are deployed across a large domestic area or internationally (Figure 4). An example of this is a multinational business that uses a WAN to interconnect their offices in different countries. The largest and best example of a WAN is the Internet, which is a network composed of many smaller networks (Figure 5). The Internet is considered the largest network in the world. The PSTN (Public Switched Telephone Network) also is an extremely large network that is converging to use Internet technologies, although not necessarily through the public Internet.

A Wide Area Network involves communication through the use of a wide range of different technologies. These technologies include Point-to-Point WANs such as Point-to-Point Protocol (PPP) and High-Level Data Link Control (HDLC), Frame Relay, ATM (Asynchronous Transfer Mode), and Sonet (Synchronous Optical Network). The difference between the WAN technologies is based on the switching capabilities they perform and the speed at which sending and receiving bits of information (data) occur.

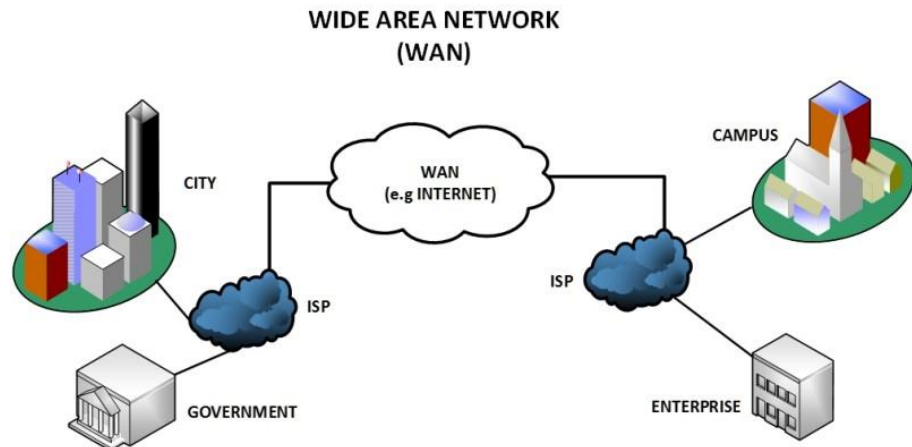


Figure 4: A Wide Area Network

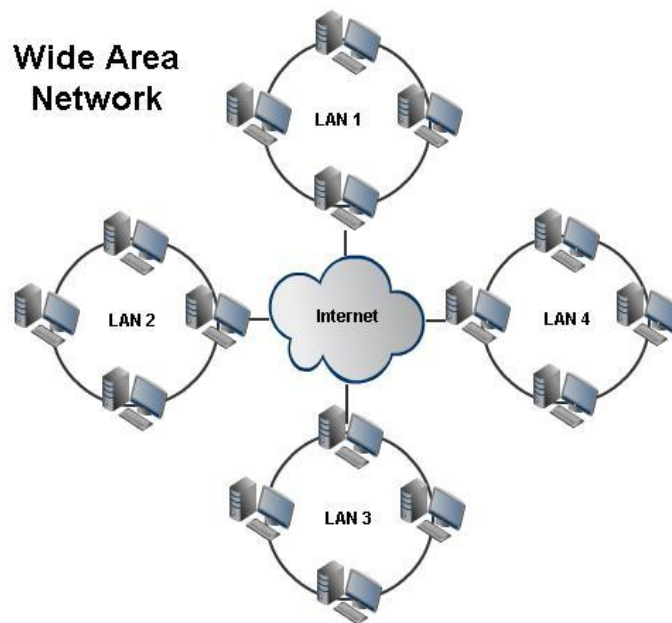


Figure 5: Internet

Wireless Networks (WLAN, WWAN): A wireless network is the same as a LAN or a WAN but there are no wires between hosts and servers (Figure 6). The data is transferred over sets of radio transceivers. These types of networks are beneficial when it is too costly or inconvenient to run the necessary cables. The media access protocols for LANs come from the IEEE. The most common IEEE 802.11 WLANs cover, depending on antennas, ranges from hundreds of meters to a few kilometers. For larger areas, either communications satellites of various types, cellular radio, or wireless local loop (IEEE 802.16) all have advantages and disadvantages. Depending on the type of mobility needed, the relevant standards may come from the IETF or the ITU.



Figure 6: Wireless Network

Network Protocol and its Components

Protocol is a set of rules that governs data communication. It represents an agreement between the communicating devices. Without protocol two devices may be connected but cannot communicate. The key elements of protocol are:

1. Syntax: Refers to the structure or format of the data, meaning the order in which they are presented.
2. Semantics: Refers how a particular pattern to be interpreted and what action is to be taken based on that interpretation.
3. Timing: Refers to when the data should be sent and how fast it should be sent.

The key functions that a protocol performs are:

- Protocol Data Unit: It refers to the breaking of data in manageable units called Protocol Data Unit e.g. Segment, Packet, Frame, etc.
- Format of packet: Format of the packet like which group of bits in the packet constitute data, address, or control bits.
- Sequencing: Refers to breaking long messages into smaller units. Sequencing is the responsibility protocol.
- Routing of packets: Finding the most efficient path between source and destination is the responsibility of protocol.
- Flow control: It is the responsibility of the protocol to prevent the fast sender to overwhelm the slow receiver. It ensures resource sharing and protection against traffic congestion by regulating the flow of data through the shared medium.
- Error control: Protocol is responsible to provide a method for error detection and correction.
- Creating and terminating a connection: Protocol defines the rules to create and terminate a connection between sender and receiver to communicate among each other.
- Security of data: Certain communication software has features to prevent data from unauthorized access.

3.4 Data Communication with Standard Telephone Lines

The public switched telephone network (PSTN) is the worldwide telephone system and usually, this network system uses digital technology. In the past, it was used for voice communication only but now it is playing very important role in data communication in the computer network such as in the Internet. There are different types of telephone lines that are used for data communication in the network. These are discussed below.

- **Dial-Up Lines:** It is a temporary connection that uses communication. Modern is used at the s telephone number is dialed from the sender the receiving end answers the call. In the communication between computers or e the cost of data communication is very Internet through this connection.
- **Dedicated Lines:** It is a permanent connection that establishes a connection between two permanently. It is better than dial-up lines connection because dedicated lines provide a constant connection. These types of connections may be digital or analog. The data transmission speed, of digital lines, is very fast as compared to analog dedicated lines. The data transmission speed is also measured in bits per second (bps). In dial-up and dedicated lines, it is up to 56 Kbps. The dedicated lines are mostly used for business purposes. The most important digital dedicated lines are described below.

Integrated Services Digital Network (ISDN)

ISDN is a set of standards used for digital transmission over the telephone line. The ISDN uses the multiplexing technique to carry three or more data signals at once through the telephone line. It is because the data transmission speed of the ISDN line is very fast. In the ISDN line, both ends of connections require the ISDN modem and a special telephone set for voice communication. Its data transmission speed is up to 128 Kbps.

Digital Subscriber Line (DSL)

In DSL, both ends of connections require the network cards and DSL modems for data communication. The data transmission speed and other functions are similar to the ISDN line. DSL transmits data on existing standard copper telephone wiring. Some DSLs provide a dial tone, which allows both voice and data communication.

Asymmetric Digital Subscriber Line (ADSL)

ADSL is another digital connection. It is faster than DSL. ADSL is much easier to install and provides a much faster data transfer rate. Its data transmission speed is from 128 Kbps up to 10 tvlbps. This connection is ideal for internet access.

Cable Television Line (CATV)

CATV line is not a standard telephone line. It is a dedicated line used to access the Internet. Its data transmission speed is 128 Kbps to 3 Mbps. A cable, modem is used with the CATV that provides a high-speed internet connection through the cable television network. A cable modem sends and receives digital data over the cable television network. To access the Internet using the CATV network, the CATV Company installs a splitter inside your house. From the splitter, one part of the cable runs to your television, and the other part connects to the cable modem. A cable modem usually is an external device, in which one end of a cable connects to a CATV wall outlet while the other end plugs into a port (such as on an Ethernet card) in the system unit.

T-Carrier Lines

It is very fast digital line that can carry multiple signals over a single communication line whereas a standard dialup telephone line carries only one signal. 1-carrier lines use multiplexing so that multiple signals share the line. T-carrier lines provide very fast data transfer rates. The T-carrier lines are very expensive and large companies can afford these

lines. The most popular T-carrier lines are:

- i. **T1 Line:** The most popular 1-carrier line is the T1 line (dedicated line). Its data transmission speed is 1.5 Mbps. Many ISPs use T1, lines to connect to the Internet backbone. Another type of TI line is the fractional TI line. It is slower than the TI line but it is less expensive. The home and business users use this line to connect to the Internet and share a connection to the T1 line with other users.

- ii. T3 Line: Another most popular and faster 1-carrier line is the T3 line. Its data transmission speed is 44 Mbps. It is more expensive than the T1 line. The main users of the T3 line are telephone companies and ISPs. The internet backbone itself also uses T3 lines.



Notes: Businesses often use T1 lines to connect to the Internet.

Asynchronous Transfer Mode (ATM)

It is a very, fast data transmission connection line that can carry data, voice, video, multimedia, etc. Telephone networks, the Internet, and other networks use ATMs. In near future, ATM will become the Internet standard for data transmission instead of T3 lines. Its data transmission speed is from 155 Mbps to 600 Mbps.

3.5 Data Communication with Modems

A modem (modulator-demodulator) is a device (Figure 7) that modulates an analog carrier signal to encode digital information, and also demodulates such a carrier signal to decode the transmitted information. The goal is to produce a signal that can be transmitted easily and decoded to reproduce the original digital data. Modems can be used over any means of transmitting analog signals, from driven diodes to radio.

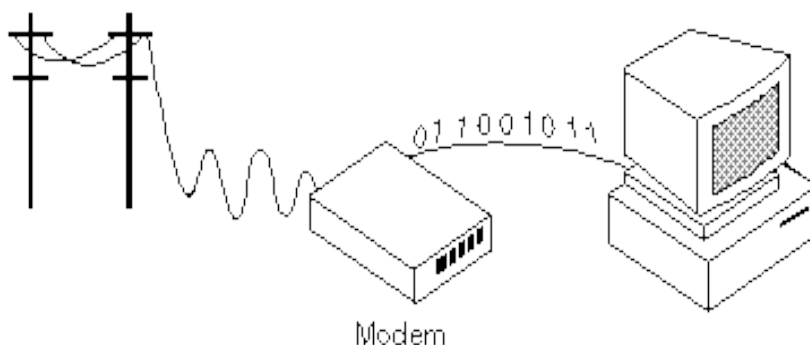


Figure 7: Data Communication with Modem

The most familiar example is a voice band modem that turns the digital data of a personal computer into modulated electrical signals in the voice frequency range of a telephone channel. These signals can be transmitted over telephone lines and demodulated by another modem at the receiver side to recover the digital data.

Modems are generally classified by the amount of data they can send in a given time unit, normally measured in bits per second (bit/s, or bps). They can also be classified by the symbol rate measured in baud, the number of times the modem changes its signal state per second. For example, the ITU V.21 standard used audio frequency-shift keying, aka tones, to carry 300 bit/s using 300 baud, whereas the original ITU V.22 standard allowed 1,200 bit/s with 600 baud using phase-shift keying.



Did You Know?

(modem) (**n.**) Short for **modulator-demodulator**. A modem is a device or program that enables a computer to transmit data over, for example, telephone or cable lines.

Fortunately, there is one standard interface for connecting external modems to computers called RS-232. Consequently, any external modem can be attached to any computer that has an RS-232 port, which almost all personal computers have. Some modems come as an expansion board that you can insert into a vacant expansion slot. These are sometimes called onboard or internal modems.

While the modem interfaces are standardized, many different protocols for formatting data to be transmitted over telephone lines exist. Some, like CCITT V.34, are official standards, while others have been developed by private companies. Most modems have built-in support for the more common protocols at slow data transmission speeds at least, most modems can communicate with each other. At high transmission speeds, however, the protocols are less standardized. Aside from the transmission protocols that they support, the following characteristics distinguish one modem from another:

Bits Per Second (BPS) : How fast the modem can transmit and receive data? At slow rates, modems are measured in terms of baud rates. The slowest rate is 300 baud (about 25 cps). At higher speeds, modems are measured in terms of bits per second (bps). The fastest modems run at 57,600 bps, although they can achieve even higher data transfer rates by compressing the data. Obviously, the faster the transmission rate, the faster you can send and receive data. Note, however, that you cannot receive data any faster than it is being sent. If, for example, the device sending data to your computer is sending it at 2,400 bps, you must receive it at 2,400 bps. It does not always pay, therefore, to have a very fast modem. Besides, some telephone lines are unable to transmit data reliably at very high rates.

Voice/data: Many modems support a switch to change between voice and data modes. In data mode, the modem acts like a regular modem. In voice mode, the modem acts like a regular telephone. Modems that support a voice/data switch have a built-in loudspeaker and microphone for voice communication.

Auto-answer: An auto-answer modem enables your computer to receive calls in your absence. This is only necessary if you are offering some type of computer service that people can call in to use.

Data compression: Some modems perform data compression, which enables them to send data at faster rates. However, the modem at the receiving end must be able to decompress the data using the same compression technique.

Flash memory: Some modems come with flash memory rather than conventional ROM, which means that the communications protocols can be easily updated if necessary.

Narrow-Band/Phone-Line Dialup Modems

The dialup modems are a standard modem of today that contain two functional parts: an analog section for generating the signals and operating the phone, and a digital section for setup and control. This functionality is often incorporated into a single chip nowadays, but the division remains in theory. In operation, the modem can be in one of two modes, data mode in which data is sent to and from the computer over the phone lines, and command mode in which the modem listens to the data from the computer for commands, and carries them out. A typical session consists of powering up the modem (often inside the computer itself) which automatically assumes command mode, then sending it the command for dialing a number. After the connection is established to the remote modem, the modem automatically goes into data mode, and the user can send and receive data.

When the user is finished, the escape sequence, “+++” followed by a pause of about a second, may be sent to the modem to return it to command mode, then a command (e.g. “ATH”) to hang up the phone is sent. Note that on many modem controllers it is possible to issue commands to disable the escape sequence so that it is not possible for data being exchanged to trigger the mode change inadvertently.

The commands themselves are typically from the Hayes command set, although that term is somewhat misleading. The original Hayes commands were useful for 300 bit/s operations only and then extended for their 1,200 bit/s modems. Faster speeds required new commands, leading to a proliferation of command sets in the early 1990s. Things became considerably more standardized in the second half of the 1990s when most modems were built from one of a very small number of chipsets. We call this the Hayes command set even today, although it has three or four times the number of commands as the actual standard.

Radio Modems

Radio modems are used extensively in direct broadcast satellite, WiFi, and mobile phones. Most of the modern telecommunications and data networks make use of them. Radio modems are used

where long-distance data links are required & formulate an important part of the PSTN. Radio modems are commonly used for high-speed computer n/w links to outlying areas where fiber is not economical.

Wireless Data Modems

The wireless data modems are used in the WiFi & WiMax standards, operating at microwave frequencies. WiFi is principally used in laptops for Internet connections (wireless access point) & wireless application protocol (WAP). They are commonly used for high-speed computer n/w links to outlying areas where fiber is not economical.

Mobile Modems and Routers

Modems that use a mobile telephone system (GPRS, UMTS, HSPA, EVDO, WiMax, etc.), are known as wireless modems (cellular modems). Wireless modems can be embedded inside a laptop or appliance or external to it. The external wireless modems connect cards, USB modems for mobile broadband, and cellular routers. A connect card is a PC card or ExpressCard (Figure 8) which slides into a PCMCIA/PC card/ExpressCard slot on a computer. The USB wireless modems (Figure 9) use a USB port on the laptop instead of a PC card or ExpressCard slot. The cellular router may have an external data card (AirCard) that slides into it.



Figure 8: T-Mobile Universal Mobile Telecommunications System PC Card Modem



Figure 9: Huawei CDMA2000 Evolution-Data Optimized USB Wireless Modem

Broadband

ADSL modems, a more recent development, are not limited to the telephone's voiceband audio frequencies. Some ADSL modems use coded orthogonal frequency division modulation (DMT, for Discrete MultiTone; also called COFDM, for digital TV in much of the world). The broadband modems use complex waveforms to carry digital data and are more advanced devices than

traditional dial-up modems as they are capable of modulating/demodulating hundreds of channels simultaneously.

Home Networking

The modems are also used for high-speed home networking applications, especially those using existing home wiring. One example is the G.hn standard, developed by ITU-T, which provides a high-speed (up to 1 Gbit/s). The LANs with such modems can be using existing home wiring (power lines, phone lines, and coaxial cables).

Deep-space Telecommunications

Many modern modems have their origin in deep space telecommunications systems of the 1960s. The comparison of deep space telecom modems with landline modems lies on following features:

- a) digital modulation formats that have high doppler immunity are typically used.
- b) waveform complexity tends to be low, typically binary phase-shift keying.
- c) error correction varies from mission to mission but is typically much stronger than most landline modems.

Voice Modems

Voice modems are regular modems that are capable of recording or playing audio over the telephone line and are mostly used for telephony applications. They can be used as an FXO card for Private branch exchange systems (compare V.92).

3.6 Data Communication Using Digital Data Connections

A communications system may be digital either by the nature of the information (also known as data) which is passed or like the signals which are transmitted. If either of these is digital then for our purposes it is considered to be a digital communications system. There are four possible combinations of data and signal types:

1. Digital data, analog signal
2. Analog data, digital signal
3. Digital data, digital signal
4. Analog data, analog signal

Digital Data with Analog Signals

This method is used to send computer information over transmission channels that require analog signals, like a fiber-optic network, computer modems, cellular phone networks, and satellite systems. In each of these systems, an electromagnetic carrier wave is used to carry the information over great distances and connect digital information users at remote locations. The digital data is used to modulate one or more of the parameters of the carrier wave. This basic process is given the name “shift-keying” to differentiate it from the purely analog systems like AM and FM.

As with analog modulation, there are three parameters of the carrier wave to vary and therefore three basic types of shift keying:

Amplitude Shift Keying (ASK): The carrier wave amplitude is changed between discrete levels (usually two) in accordance with digital data. Here, the digital data to be transmitted is binary number 1011. Two amplitudes are used to directly represent the data, either 0 or 1. This modulation is called binary amplitude shift keying or BASK.

Frequency Shift Keying (FSK): The carrier frequency is changed between discrete values. If only two frequencies are used then this will be called BFSK, for binary frequency shift keying.

Phase Shift Keying (PSK): The phase of the carrier wave at the beginning of the pulse is changed between discrete values. We have *amplitude phase keying* that uses combinations of amplitude and phase keying. For example, if we use two levels of amplitude and two levels of phase together, there will be a total of four possibilities. This is used to transmit two independent channels of digital data simultaneously. It is also called Quadrature AM or Quaternary PSK. Another type of phase keying is *Minimum Shift Keying (MSK)* that is used to find the minimum signal bandwidth for a particular method (usually FSK).

Analog Data with Digital Signals

A digital signal can be transmitted over a dedicated connection between two or more users. In order to transmit analog data, it must first be converted into a digital form. This process is called sampling, or encoding.

Sampling involves two steps, that is, taking the measurements at regular sampling intervals and then converting the value of the measurement into binary code. Sampling has certain considerations where the amplitude of a signal is measured at regular intervals and the interval is designated as t_s and is called sample interval. However, the sample interval must be chosen to be short enough. The sampling rate (inverse of the sample interval) should be greater than twice the highest frequency component of the signal which is being sampled. Known as the Nyquist frequency. If sampling is done at a lower rate, then there is the risk of missing some information, known as aliasing.

Encoding is the process where once the samples are obtained, they must be encoded into binary. For a given number of bits, each sample may take on only a finite number of values. If more bits are used for each sample, then a higher degree of resolution is obtained. The higher bit sampling requires more storage for data and requires more bandwidth to transmit.

Digital Data with Digital Signals

Digital data is a collection of zeroes and ones, where the value at any one time is called a bit. There are several different ways in which a binary number be formatted. This is called pulse code modulation or PCM. The most straightforward PCM format is designated as NRZ-L (nonreturn to zero levels). In this format, the level directly represents the binary value: low level = 0, high level = 1.

Analog Data with Analog Signals

When a low-pass analog signal is converted into a bandpass analog signal, it is called analog-to-analog conversion. It is a similar situation when data from one computer is sent to another via some analog carrier, it is first converted into analog signals.

3.7 Wireless Networks

A wireless network refers to any type of computer network that is not connected by cables of any kind. It is a method by which telecommunications networks and enterprise (business), installations avoid the costly process of introducing cables into a building, or as a connection between various equipment locations. Wireless telecommunications networks are generally implemented and administered using a transmission system called radio waves. This implementation takes place at the physical level, (layer), of the network structure.

Types of Wireless Connections

Wireless Local Area Networks (WLANs): WLANs allow the users in a local area, such as a university campus or library, to form a network or gain access to the internet (Figure 10). These are temporary networks that can be formed by a small number of users without the need for an access point.

Wireless Personal Area Networks (WPANs): A personal telecommunications device that communicates in the individual distance (Figure 11). The two current technologies include Infra-Red (IR) and Bluetooth (IEEE 802.15). WPANs allow the connectivity of personal devices within an area of about 30 feet. The IR requires a direct line of sight and the range is less.

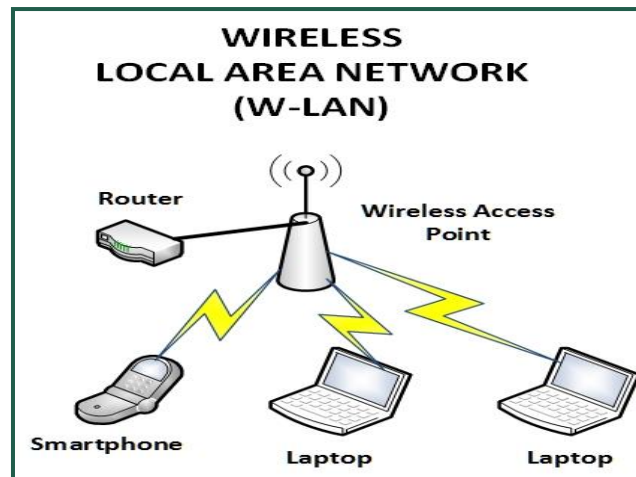


Figure 10: Wireless LANs



Figure 11: WPANs Scenario

Wireless Metropolitan Area Networks (WMANs): The WMANs allow the connection of multiple networks in a metropolitan area such as different buildings in a city as depicted in Figure 12.

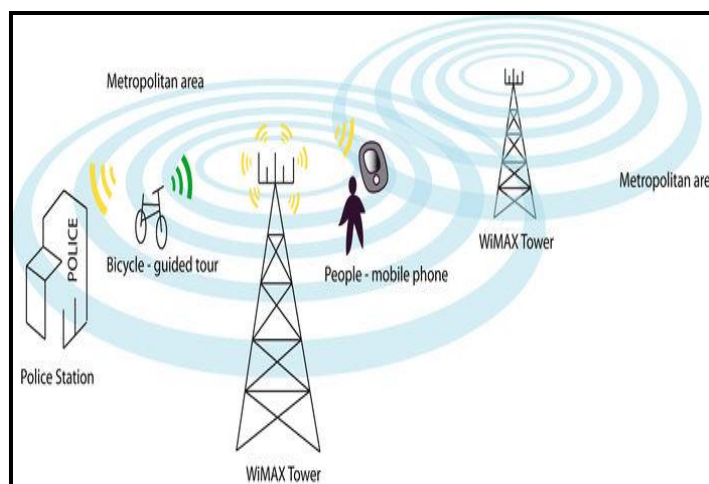


Figure 12: WMANs Scenario

Wireless Wide Area Networks (WWANs): These can be maintained over large areas, such as cities or countries, via multiple satellite systems or antenna sites looked after by an ISP. Currently, Information and Communication Technology (ICT) sector is focusing on 5G (5th generation)

systems. WWAN connect two or more areas in geographical context e.g. state, province, country such as Notre Dame in Fremantle and Broome (Figure 13).

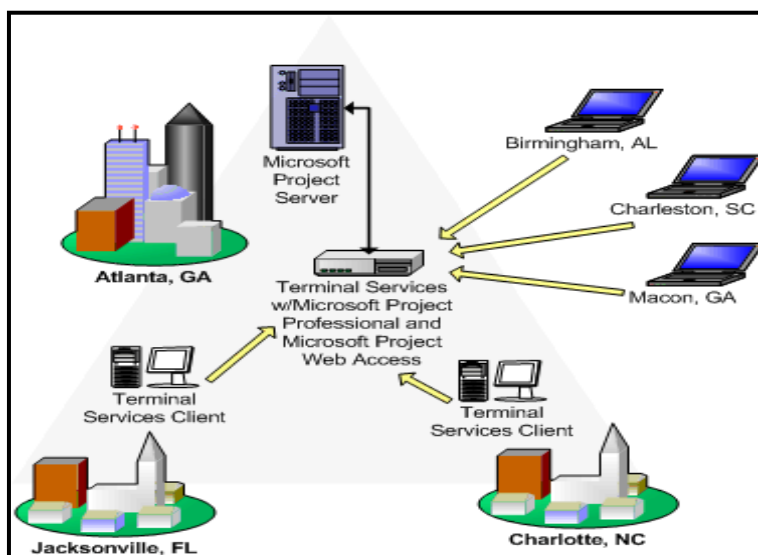


Figure 13: Wireless Wide Area Networks (WWANs)

Difference Between Different Types of Wireless Connections

Type	Coverage	Performance	Standards	Applications
Wireless PAN	Within reach of a person	Moderate	Wireless PAN Within reach of a person Moderate Bluetooth, IEEE 802.15, and IrDa Cable replacement for peripherals	Cable replacement for peripherals
Wireless LAN	Within a building or campus	High	IEEE 802.11, Wi-Fi, and HiperLAN	Mobile extension of wired networks
Wireless MAN	Within a city	High	Proprietary, IEEE 802.16, and WIMAX	Fixed wireless between homes and businesses and the Internet
Wireless WAN	Worldwide	Low	CDPD and Cellular 2G, 2.5G, and 3G	Mobile access to the Internet from outdoor areas

Summary

- Digital communication is the physical transfer of data over a point to point or point to a multipoint communication channel.
- The public switched telephone network (PSTN) is a worldwide telephone system and usually, this network system uses the digital technology.
- A modem is a device that modulates an analog carrier signal to encode, digital information, **Notes** and also demodulates such a carrier signal to decode the transmitted information.

- A wireless network refers to any type of computer network that is not connected by cables of any kind.
- Wireless telecommunication networks are generally implemented and administered using a transmission system called radio waves.

Keywords

Computer networking: A computer network, often simply referred to as a network, is a collection of computers and devices interconnected by communications channels that facilitate communications among users and allows users to share resources. Networks may be classified according to a wide variety of characteristics.

Data transmission: Data transmission, digital transmission, or digital communications is the physical transfer of data (a digital bitstream) over a point-to-point or point-to-multipoint communication channel.

Dial-Up lines: Dial-up networking is an important connection method for remote and mobile users. A dial-up line is a connection or circuit between two sites through a switched telephone network.

DNS: The Domain Name System (DNS) is a hierarchical naming system built on a distributed database for computers, services, or any resource connected to the Internet or a private network.

DSL: Digital Subscriber Line (DSL) is a family of technologies that provides digital data transmission over the wires of a local telephone network.

GSM: Global System for Mobile Communications, or GSM (originally from Groupe Spécial Mobile), is the world's most popular standard for mobile telephone systems.

ISDN Lines: Integrated Services Digital Network (ISDN) is a set of communications standards for simultaneous digital transmission of voice, video, data, and other network services over the traditional circuits of the public switched telephone network.

LAN: A local area network (LAN) is a computer network that connects computers and devices in a limited geographical area such as a home, school, computer laboratory, or office building.

MAN: A metropolitan area network (MAN) is a computer network that usually spans a city or a large campus.

Modem: A modem (modulator-demodulator) is a device that modulates an analog carrier signal to encode digital information, and also demodulates such a carrier signal to decode the transmitted information.

PSTN: The public switched telephone network (PSTN) is the network of the world's public circuit-switched telephone networks. It consists of telephone lines, fiber optic cables, microwave transmission links, cellular networks, communications satellites, and undersea telephone cables all inter-connected by switching centers that allow any telephone in the world to communicate with any other.

WAN: A wide area network (WAN) is a computer network that covers a broad area (i.e., any network whose communications links cross metropolitan, regional, or national boundaries).

WISP: Wireless Internet Service Providers (WISPs) are Internet service providers with networks built around wireless networking.

Self Assessment

1. What is Data Communication?
 - A. The transmission of data from one location to another for direct use or further processing.
 - B. A system made up of hardware, software and communication facilities, computer terminals, and input\output devices linked locally or globally.
 - C. A telegraphy system.
 - D. A channel is used to transmit data at a rate of 1000-8000cps.

2. The three classes of channel that makes up bandwidth are:
- A. Data, communication, transmission.
 - B. Analogue signals, digital signals, standard telephone lines.
 - C. Narrow-band, voice-band, broad-band channels.
 - D. Automatic dialing, signal transmission, modems.
3. A data communication system consists of
- A. The transmission of data from one location to another for indirect use.
 - B. A system made up of hardware, software and communication facilities, computer terminals, and input\output devices linked locally or globally.
 - C. A telegraphy system.
 - D. A channel used to transmit data at a rate of 1000-8000cps.
4. Digital communication is the _____ transfer of data over a point to point or point to multipoint communication channel.
- A. logical
 - B. virtual
 - C. physical
 - D. singular
5. A bandwidth determines the volume of data that can be transmitted in a given time?
- A. True
 - B. False
6. LAN is a computer network that connects computers and devices in a _____.
- A. limited geographical area
 - B. expanded geographical area
 - C. distributed manner across multiple domains
 - D. metropolitan area
7. A modem is a device that _____ an analog carrier signal to encode digital information, and also _____ a carrier signal to decode the transmitted information.
- A. carries, observes
 - B. observes, carries
 - C. demodulates, modulates
 - D. modulates, demodulates
8. A _____ is a computer network that usually spans a city or a large campus.
- A. Local Area Network (LAN)
 - B. Metropolitan Area Network (MAN)
 - C. Global Area Network (GAN)
 - D. City Area Network (CAN)
9. IEEE 802.11, Wi-Fi, and HiperLAN are the standards for _____.
- A. Wireless PAN
 - B. Wireless MAN

- C. Wireless WAN
- D. Wireless LAN

10. WISP stands for:

- A. Wireless Instantaneous Source Provider
- B. Wireless Internet Service Providers
- C. Wideband Internet Service Providers
- D. Wideband Internet Source Protocol

11. A digital signal can be transmitted over a dedicated connection between two or more users. In order to transmit analog data, it must first be converted into a digital form. This process is called as _____.

- A. Demodulation
- B. Modulation
- C. Sampling
- D. Transmission

12. The data transmission speed for _____ ranges from 155 Mbps to 600 Mbps.

- A. Asymmetric Digital Subscriber Line (ADSL)
- B. Cable Television Line (CATV)
- C. Asynchronous Transfer Mode (ATM)
- D. T-Carrier Lines

13. ASK is a result of a combination of shift keying and

- A. digital modulation
- B. amplitude modulation
- C. analog modulation
- D. none of the above

14. Which of the following is not a digital-to-analog conversion?

- A. ASK
- B. PSK
- C. FSK
- D. AM

15. Integrated Services Digital Network uses the _____ technique to carry three or more data signals at once through the telephone line.

- A. demultiplexing
- B. encoding
- C. aliasing
- D. multiplexing

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. B | 4. C | 5. A |
| 6. A | 7. D | 8. B | 9. D | 10. B |
| 11. C | 12. C | 13. B | 14. D | 15. D |

Review Questions

1. What do you mean by data communication?
2. Explain the general model of data communication. What is the role of the modem in it?
3. Explain the general model of digital transmission of data. Why is analog data sampled?
4. What do you mean by digital modulation? Explain various digital modulation techniques.
5. What are computer networks?
6. How data communication is done using standard telephone lines?
7. What is ATM switch? Under what condition it is used?
8. What do you understand by ISDN?
9. What are the different network methods? Give a brief introduction about each.
10. What do you understand by wireless networks? What is the use of the wireless network?
11. Give the types of wireless networks.
12. What is the difference between broadcast and point-to-point networks?

**Further Reading**

Computing Fundamentals by Peter Norton.

**Web Links**

[Data Communication & Computer Network - Tutorialspoint](#)

[Data Communication - What is Data Communication? - Computer Notes \(ecomputernotes.com\)](#)